

SIMPLE GROUPS AND COMPLEMENTS OF SMOOTH SURFACES IN SIMPLY CONNECTED 4-MANIFOLDS

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ABSTRACT. For each integer n we construct an oriented closed simply connected 4-manifold X admitting a smoothly embedded closed connected surface Σ of self-intersection number n such that the complement of the surface has non-trivial fundamental group. This answers a question of Kronheimer in Kirby's 1997 problem list. The proof combines a topological construction with homological properties of simple groups such as Thompson's group V and certain sporadic finite simple groups.

1. INTRODUCTION

A basic invariant of an embedding of a surface Σ in a 4-manifold X is the fundamental group of its complement, and it is reasonable to ask what groups can occur, especially if X is simply connected. With no further restrictions on the embedding, there is a simple characterisation of such groups [KR08]: A finitely presented group G is $\pi_1(X - \Sigma)$ for a connected surface Σ in some X (not specified in advance) if and only if $H_1(G)$ is cyclic and G is the normal closure of a single element; a question of Wiegold [KM23, Problem 5.52] asks if the second condition follows automatically from the first.

The construction in [KR08] provides a surface (in fact a symplectically embedded surface in a symplectic 4-manifold) with self-intersection 0, but the problem is more challenging if one requires that the self-intersection is non-zero. Indeed, a question of Kronheimer in Kirby's 1997 problem list [Kir97, Problem 4.109] asks, for simply connected X , whether $\pi_1(X - \Sigma)$ is trivial if Σ has non-zero self-intersection that is square-free (all prime divisors appear once). The motivation is an observation of Kronheimer–Mrowka [KM93, Proposition 5.7] that in this situation, $\pi_1(X - \Sigma)$ would have no non-trivial representations in $\mathrm{SO}(3)$ or $\mathrm{SU}(2)$. This fact suggested that possible fundamental groups of surface complements might be found among groups without unitary representations, and led to the consideration of Thompson's group V , discussed in Section 3.

In this note, we construct surfaces with non-simply connected complements and provide a negative answer to Kronheimer's question. The result applies to all non-zero self-intersections, not just square-free ones.

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Theorem A. *For any $n \in \mathbb{Z}$, there is an oriented closed simply connected 4-manifold X and surface Σ smoothly embedded in X such that $\Sigma \cdot \Sigma = n$ and $\pi_1(X - \Sigma) \neq \{1\}$. If n is even we may take X to be spin.*

Remark 1.1. We note here some topological preliminaries.

- (i) All surfaces and 4-manifolds discussed in the paper will be assumed compact and connected.
- (ii) The sign of the self-intersection $\Sigma \cdot \Sigma$ depends on a choice of orientation for X which we make implicitly throughout. Note that by reversing the orientation of X , the statement of Theorem A for n is equivalent to the same statement for $-n$.
- (iii) The condition that the self-intersection of $\Sigma \subset X$ be square-free implies that the homology class $[\Sigma] \in H_2(X)$ is primitive. More generally, if $[\Sigma] = dA$ where A is a primitive class in $H_2(X; \mathbb{Z})$, then we call d the multiplicity of Σ . As in [HS71, Lemma 3.1], if $[\Sigma]$ has multiplicity d , then $H_1(X - \Sigma) \cong \mathbb{Z}/d$ and $\Sigma \cdot \Sigma = d^2 A \cdot A$.

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2. FROM GROUP THEORY TO SURFACE COMPLEMENTS

Let Σ be a closed orientable surface of genus at least one. Choose an orientation for Σ , and let N be the disk bundle over Σ of Euler class n with orientable total space. Its boundary, a circle bundle over Σ , will be denoted by Y . We note that for an embedding of Σ with normal bundle N , the complement $X - \Sigma$ of Σ deformation retracts onto the exterior of Σ , defined as $X - \text{int}(N)$; it is more convenient henceforth to work with the exterior. The idea of the proof of Theorem A is to find a suitable manifold W with $\partial W = Y$ to serve as the exterior of Σ in X . In fact we will take X to be $N \cup W$, and the goal is to choose W so that X is simply connected.

To this end, we make use of the oriented bordism group $\Omega_k(BG)$, where BG is the classifying space for the group G ; see [Ati61], where $\Omega_k(\cdot)$ is denoted $\text{MSO}_k(\cdot)$. The addition in this group is given by disjoint union. The elements of this group are equivalence classes of pairs $(M, f: M \rightarrow BG)$, where M is a compact oriented k -manifold, and (M, f) and (M', f') are equivalent if they are cobordant via an oriented manifold V and a map $F: V \rightarrow BG$ extending f and f' . Note that if M is a connected oriented closed k -manifold, a homomorphism $\varphi: \pi_1(M) \rightarrow G$ is induced by a map $M \rightarrow BG$, well-defined up to homotopy. The bordism class of this map will be denoted $[M, \varphi] \in \Omega_k(BG)$; if M is disconnected, then φ would be a collection of homomorphisms of the fundamental groups of its components, and one simply adds the bordism elements. If there is an oriented cobordism V between M and M' and a homomorphism from $\pi_1(V) \rightarrow G$ restricting to φ and φ' on M and M' , respectively, then $[M, \varphi] = [M', \varphi']$. One defines,

similarly, the spin bordism group $\Omega_k^{\text{spin}}(BG)$ as spin cobordism classes of triples $(M, \mathfrak{s}, f)$ where $\mathfrak{s}$ is a spin structure on M and $f: M \rightarrow BG$ as above.

The following lemma is a straightforward consequence of the Atiyah-Hirzebruch spectral sequences

$$E_{p,q}^2 = H_p(X; \Omega_q(pt)) \quad \text{and} \quad E_{p,q}^2 = H_p(X; \Omega_q^{\text{spin}}(pt))$$

for the generalised homology theory of oriented bordism [AH61, McC01] or spin bordism. The first part can be proved geometrically as in [Gor86]. Note [Bro94, §I.6] that the group homology $H_*(G)$ is the same as the homology $H_*(BG)$ of the classifying space; we will use the former notation as it is more standard in the group theory literature.

Lemma 2.1. *The map $\Omega_3(BG) \rightarrow H_3(G; \mathbb{Z})$ that assigns $\varphi_*([M])$ to a pair $[M, \varphi]$ is an isomorphism. If $H_1(G; \mathbb{Z}) = H_2(G; \mathbb{Z}) = 0$, then the same is true for $\Omega_3^{\text{spin}}(BG)$.*

Fix an orientation of the fibres of Y . Recall that the fundamental group of Y is a central extension

$$\{1\} \longrightarrow \mathbb{Z} \longrightarrow \pi_1(Y) \longrightarrow \pi_1(\Sigma) \longrightarrow \{1\}$$

and that the $\mathbb{Z}$ subgroup is the kernel of the map $\pi_1(Y) \rightarrow \pi_1(N)$. Denote by μ the generator of this $\mathbb{Z}$ subgroup.

Lemma 2.2. *Suppose that there is a finitely presented perfect group P and a homomorphism $\varphi: \pi_1(Y) \rightarrow P$ such that $\varphi(\mu)$ normally generates P and the bordism class $[Y, \varphi] \in \Omega_3(BP)$ is trivial. Then there is an embedding of N in a simply connected 4-manifold X such that the exterior $W = X - \text{int}(N)$ has fundamental group P . If n is even, choose a spin structure on N with restriction σ_Y to Y . If the bordism class of (Y, σ_Y, φ) in $\Omega_3^{\text{spin}}(BP)$ is trivial, then X may be chosen to be a spin manifold.*

Proof. By hypothesis, there is a compact 4-manifold W_0 with boundary Y and the portion connected by solid arrows in the diagram below.

$$\begin{array}{ccccc} \pi_1(Y) & \xrightarrow{\varphi} & P & & \\ & \searrow j_* & \nearrow \Phi_0 & \nearrow \Phi_1 & \nearrow \Phi \\ & & \pi_1(W_0) & \xrightarrow{i} & \pi_1(W_1) & \xrightarrow{q} & \pi_1(W) \end{array}$$

We modify W_0 in two stages to fill in the dotted portion of the diagram so that Φ is an isomorphism. Since P is finitely presented, there is a closed 4-manifold W_P with $\pi_1(W_P) \cong P$. Replace W_0 with $W_0 \# W_P$ and note that the homomorphism $\pi_1(W_0 \# W_P) \cong \pi_1(W_0) * P \rightarrow P$ given by Φ_0 on the first factor and the identity on the second factor is a surjection. Choose finitely many generators $\{x_j\}$ for $\pi_1(W_0)$. Now do surgery on circles in $W_0 \# W_P$ representing the elements $x_j^{-1}\Phi_0(x_j)$ to obtain a manifold W_1 with a surjection Φ_1 as indicated. The homomorphism $i: \pi_1(W_0) \rightarrow \pi_1(W_1)$ is induced by the inclusion of the summand W_0 in $W_0 \# W_P$. As in [Wal66, Wal99, Theorem 1.4] do surgery on finitely many circles in W_1 to kill the kernel of Φ_1 , obtaining the manifold W and isomorphism Φ as indicated.

The image of μ in $\pi_1(W)$ is taken to $\varphi(\mu)$, which is assumed to be non-trivial in P . By hypothesis, $\varphi(\mu)$ normally generates $\pi_1(W)$. On the other hand, μ is trivial in $\pi_1(N)$, so van Kampen's theorem says that $X = W \cup N$ is simply connected.

When n is even, so that N has a spin structure, we describe how the argument goes through to produce a simply connected spin manifold X . By hypothesis, the initial manifold W_0 has a spin structure extending the spin structure on Y that it acquires as the boundary of N . We need to see that the surgeries done in the preceding argument can be chosen so that at each stage, the new manifold inherits a spin structure. A more general statement can be found in the surgery-theory literature, for example in [Kre99, Lemma 2], but this might be challenging for the reader. So we provide a direct proof (valid in any dimension greater than 2) of exactly the statement we need.

A normal framing of a circle γ embedded in an n -manifold Z^n is equivalent (up to isotopy) to an embedding $\varphi: S^1 \times D^{n-1} \rightarrow Z$ such that $\varphi(S^1 \times 0) = \gamma$. Write Z_φ for the result of surgery using this framing. Then there is a cobordism C_φ from Z to Z_φ obtained by adding a 2-handle $D^2 \times D^{n-1}$ to $Z \times I$ along $\varphi(S^1 \times D^{n-1}) \times \{1\}$.

Claim: Let Z^n be a spin manifold with $n \geq 3$ and let γ be a circle embedded in Z . Then there is a framing φ of γ so that Z_φ has a spin structure extending the spin structure on the exterior of γ .

Proof of Claim: Recall that the set of spin structures on a manifold M has a free and transitive action of the group $H^1(M; \mathbb{Z}/2)$. In particular, $S^1 \times D^{n-1}$ has two spin structures, one of which (the 'bounding spin structure') is the restriction of the unique spin structure on $D^2 \times D^{n-1}$, and the other of which does not extend over $D^2 \times D^{n-1}$.

Starting with one framing φ , a second framing is obtained by composing φ with the self-diffeomorphism τ of $S^1 \times D^{n-1}$ built from the non-trivial element of $\pi_1(\mathrm{SO}(n-1))$. It is straightforward to show that the pull-back of the bounding spin structure on $S^1 \times D^{n-1}$ is the non-bounding one, and vice versa. Now consider a spin structure σ_Z on Z that we'd like to extend over the cobordism C_φ . If the pull-back of σ_Z to $S^1 \times D^{n-1}$ is the bounding one, then σ_Z extends over the cobordism C_φ , yielding the desired spin structure on Z_φ . If, on the other hand, $\varphi^*\sigma_Z$ is the non-bounding spin structure, the spin structure $(\varphi \circ \tau)^*\sigma_Z$ is the bounding one, yielding a spin structure on $Z_{\varphi \circ \tau}$.

This establishes the claim, and completes the proof of the lemma. $\square$

As remarked at the top of this section, the embedding of N in X yields an embedding of Σ for which the complements have the same fundamental group. Hence, to complete the proof of Theorem A, it suffices to find groups and homomorphisms satisfying the hypotheses of Lemma 2.2; this is carried out in the next two sections of the paper. The first of these makes use of Thompson's infinite simple group, while the second uses an assortment of finite simple groups.

3. FROM CIRCLE BUNDLES TO THOMPSON'S GROUP

A group G is *type* F_n if it admits a $K(G, 1)$ with finite n -skeleton and *type* F_∞ if it is *type* F_n for all $n \geq 0$. A group G is *type* FP_n if it admits a projective resolution $P_\bullet \rightarrow \mathbb{Z}$ over $\mathbb{Z}G$ with P_i finitely generated for $i \leq n$ and *type* FL_n if the P_i can be taken to be finitely generated free modules for $i \leq n$. We say G is *type* FP_∞ if it is *type* FP_n for all n and *type* FL_∞ if it is *type* FL_n for all n . Note that FP_n and FL_n are equivalent [Bro94, VIII, Prop 4.1] and that Wall proved in [Wal65] that a finitely presented group of *type* FL_n is *type* F_n .

Proposition 3.1. *Let $S^1 \rightarrow Y \rightarrow \Sigma_g$ exhibit Y as a circle bundle over a surface of genus $g \geq 1$. Let $H = \pi_1(Y)$ and let $\mu \in H$ be a generator of $\pi_1(S^1) < H$. Then, there exists a group V and a homomorphism $\psi: H \rightarrow V$ such that*

- (1) V is finitely presented, in fact F_∞ ;
- (2) $H_i(V; \mathbb{Z}) = 0$ for $i = 1, 2, 3$; in particular $\Omega_3(BV) = \Omega_3^{\text{spin}}(BV) = 0$;
- (3) $\psi(\mu)$ normally generates V .

Proof. We let V denote Thompson's group V (see [Hig74] for a definition). We recount the following facts:

- (1) V is a simple group [Hig74];
- (2) V is acyclic [SW19], so $H_i(V; \mathbb{Z}) = 0$ for all $i \geq 1$;
- (3) V contains every finite group as a subgroup [Hig74];
- (4) V is finitely presented [Hig74];
- (5) V is *type* FP_∞ [Bro87];
- (6) V is *type* F_∞ , which follows from the previous two items.

By [Hem87] the group H is residually finite. Thus, we can find a finite quotient $h: H \rightarrow Q$ such that $h(\mu)$ is non-trivial. Since every finite group is a subgroup of V we may embed Q into V via some homomorphism $i: Q \rightarrow V$. We define ψ to be $i \circ h$. Now, as V is simple, every non-trivial element of V normally generates it. Thus, $\psi(\mu)$ normally generates V . $\square$

Remark 3.2. When the Euler number of the circle bundle n is greater than one in absolute value, we do not have to appeal to the residual finiteness of H . For in this case, H has a surjection onto $\mathbb{Z}/n$ taking μ to a generator. By item (3) above, $\mathbb{Z}/n$ is a subgroup of V , which provides the desired ψ . It is slightly more delicate to find explicit homomorphisms when $n = \pm 1$; in the next section we do this with V replaced by various finite simple groups.

Proof of Theorem A. Combining Proposition 3.1 with Lemma 2.2 completes the proof of Theorem A. Here the complements $X - \Sigma$ have fundamental group V . $\square$

Remark 3.3 (Higman–Thompson groups). In fact we can build more examples out of the Higman–Thompson groups $V_{m,r}$. In [SW19], the authors show $H_*(V_{m,r}; \mathbb{Z}) \cong H_*(\Omega_0^\infty \mathbf{M}_{m-1}; \mathbb{Z})$, where the second object is the homology of the zeroth component of the infinite loop space of the mod $m-1$ Moore spectrum. The relevance for us is Propositions 6.1 and 6.2 of *ibid*; there it is shown that when m is even and p is the smallest prime dividing $m-1$ we have $\tilde{H}_d(V_{m,r}; \mathbb{Z}) = 0$ for $d < 2p-3$ and $H_{2p-3}(V_{m,r}; \mathbb{Z}) = \mathbb{Z}/p$.

In particular, these groups can be realised as the fundamental group of a surface complement with one caveat: if $3|(m-1)$, then one must map to a finite group Q with $|H_3(Q; \mathbb{Z})|$ coprime to 3 to ensure the bordism class vanishes.

Remark 3.4. Higman–Thompson groups have no non-trivial linear representations. Indeed, by Mal’cev’s Theorem [Mal65], a finitely generated linear group is residually finite. Hence, any finitely generated group admitting a non-trivial linear representation also admits a non-trivial finite quotient. But V_n is simple and so does not admit any non-trivial finite quotients.

4. FINITE SIMPLE GROUPS

In this section, to demonstrate the flexibility of the method, we construct examples of surface complements with fundamental group isomorphic to certain finite simple groups. This demonstrates that the fundamental group of a surface complement need not be trivial or infinite. It is also noteworthy because finite groups have many linear representations, as opposed to the Higman–Thompson groups, which have no non-trivial linear representations (see Remark 3.4). Note that the finite groups we consider have no non-trivial representations in $\mathrm{SO}(3)$ or $\mathrm{SU}(2)$, so we do not contradict [KM93, Proposition 5.7].

Self-intersection number one. We first consider the case where the disk bundle over the surface Σ has Euler number equal to one or minus one.

This section will use a number of notions from finite group theory so before we begin we establish some notation. The group $\mathrm{He}_3(p)$ is the (3×3) -Heisenberg group over $\mathbb{F}_p$ and is isomorphic to the group of (3×3) -uni-triangular matrices over the finite field $\mathbb{F}_p$ of p -elements. Here a uni-triangular matrix is upper triangular with ones on the diagonal. We will refer to a number of sporadic simple groups: the Mathieu groups M_{22} and M_{23} , the Higman–Sims group HS , the Held group He , the McLaughlin group McL , the Janko groups J_3 and J_4 , the Lyons group Ly , the Conway groups Co_3 , Co_2 , and Co_1 , the Suzuki group Suz , the O’Nan group $O’N$, the Fisher group Fi_{22} and the Monster group M . Our notation for these groups is fairly standard and follows the Atlas of finite simple groups [CCN⁺85].

Proposition 4.1. *Let Σ be a closed oriented surface of genus at least one. Each sporadic simple group in Table 1 can be realised as the fundamental group of a surface complement $X - \Sigma$ with $\Sigma \cdot \Sigma = \pm 1$, where X is some closed oriented simply connected 4-manifold.*

Proof. Let H denote the fundamental group of the associated circle bundle over Σ . Since $\Sigma \cdot \Sigma = \pm 1$, H has a presentation

$$\left\langle a_1, b_1, \dots, a_g, b_g, z \mid \prod_{i=1}^g [a_i, b_i] = z, [a_j, z] = [b_j, z] = 1 \text{ for } 1 \leq j \leq g \right\rangle.$$

The group H admits a surjection onto $\mathrm{He}_3(\mathbb{Z})$, the 3-dimensional integer Heisenberg group (uni-triangular matrices), with presentation

$$\langle a, b, z \mid [a, b] = z, [a, z] = [b, z] = 1 \rangle.$$

	M_{22}	M_{23}	HS	He	McL	J_3	J_4	Ly
$H_2(P; \mathbb{Z})$	12	0	2	0	0	3	0	0
$H_3(P; \mathbb{Z})$	0	0	2+2	12	0	15	0	0
$He_3(p)$	2	2	5	7	2,3,5	2	2,3	2,3,5
	Co_3	Co_2	Co_1	Suz	$O'N$	Fi_{22}	M	
$H_2(P; \mathbb{Z})$	0	0	2	6	3	6	0	
$H_3(P; \mathbb{Z})$	6	4	12	4	8	1	24 + (≤ 4)	
$He_3(p)$	5	3,5	5	3	7	2,3	5, 7	

TABLE 1. Some sporadic simple groups P where the third homology is known (except in the case of the Monster group M where there are three possibilities) and which admit a $He_3(p)$ subgroup with p coprime to the order of the third homology of P .

The notation $m + n$ in the homology rows is interpreted as $\mathbb{Z}/m \oplus \mathbb{Z}/n$; $a \leq k$ means a subgroup of $\mathbb{Z}/k$ appears a summand.

The p for which such a subgroup exists are listed in the $He_3(p)$ row.

The homomorphism is given by $a_1 \mapsto a$, $b_1 \mapsto b$, $a_i, b_i \mapsto 1$ for $i \geq 2$, and of course $z \mapsto z$. For each prime p , the group $He_3(\mathbb{Z})$ admits a surjection onto the p -group $He_3(p)$ by considering the modulo p reduction of matrices in $He_3(\mathbb{Z})$. Note that when $p = 2$ this group is the dihedral group on 4 points D_4 containing 8 elements.

We want to apply Lemma 2.2; the following conditions are sufficient on some finite simple group P :

- (1) $He_3(p) \leq P$
- (2) $H_3(P; \mathbb{Z})$ has no p -torsion.

Indeed, we want the class (Y, φ) to be trivial in $\Omega_3(BP)$, but this class is in the image of the composition

$$H_3(Y; \mathbb{Z}) \rightarrow H_3(\pi_1(Y); \mathbb{Z}) \rightarrow H_3(He_3(p); \mathbb{Z}) \rightarrow H_3(P; \mathbb{Z})$$

and the group $H_3(He_3(p); \mathbb{Z})$ is annihilated by multiplication by p^3 . So the last map is trivial and hence $[Y, \varphi] = 0 \in \Omega_3(BP)$. Thus, to complete the proof we just have to confirm the veracity of the information in Table 1.

In Table 1, the existence or non-existence of the $He_3(p)$ subgroups is easily verified by consulting the ATLAS of finite simple groups [CCN⁺85]. The homology computations were obtained from the following sources: the low-dimensional homology of the Mathieu groups can be found in [DSE09] and the remaining computations are contained in [JFT19]. Note that for the Monster group M , it is known that $H_3(M; \mathbb{Z})$ is a subgroup of $\mathbb{Z}/24 \oplus \mathbb{Z}/4$ but the exact group is not known. The groups M_{11} , M_{12} , M_{24} , J_1 , J_2 have been omitted from Table 1 due to having no $He_3(p)$ subgroup with p coprime to $|H_3(P; \mathbb{Z})|$. The groups Ru , HN , Th , Fi_{23} , Fi'_{24} , and B have been omitted from Table 1 because neither $H_3(G; \mathbb{Z})$ nor its prime divisors are known. $\square$

Self-intersection number greater than one. As remarked in Section 3, when the self-intersection number n satisfies $|n| > 1$ there is a surjection $\pi_1(Y) \twoheadrightarrow \mathbb{Z}/n$ such that the image of μ is a generator of $\mathbb{Z}/n$. Here we will

show that for n divisible by a prime $p \geq 7$, we can realise the simple group $\mathrm{PSL}_2(p)$ as a surface complement.

Recall that $\mathrm{SL}_2(p)$ is the group of invertible 2×2 -matrices over $\mathbb{F}_p$ with determinant equal to 1 and that $Z(\mathrm{SL}_2(p)) = \{\pm I_2\}$. We identify $\mathrm{PSL}_2(p)$ with $\mathrm{SL}_2(p)/Z(\mathrm{SL}_2(p))$.

Proposition 4.2. *For each prime $p \geq 7$, integer $g \geq 0$, and $n \in \mathbb{Z}$ such that $p|n$, the group $\mathrm{PSL}_2(p)$ can be realised as the fundamental group of a surface complement $X - \Sigma$, where X is a closed oriented simply connected 4-manifold, Σ has genus g , and $\Sigma \cdot \Sigma = \pm n$.*

Proof. The key point is that $\mathbb{Z}/n$ surjects onto $\mathbb{Z}/p$ and this latter group is isomorphic to a Sylow p -subgroup of the simple group $P = \mathrm{PSL}_2(p)$. Before we prove the proposition we show that $H_3(P; \mathbb{Z})$ has no elements of order p .

Note that a Sylow p -subgroup Q of P is isomorphic to $\mathbb{Z}/p$. We may take Q to be the image of the upper triangular matrices with 1s on the diagonal under the projection $\mathrm{SL}_2(p) \rightarrow \mathrm{PSL}_2(p)$. The action of $N_P(Q)$ on Q factors through a homomorphism $\phi : N_P(Q) \rightarrow \mathrm{Aut}(Q)$ whose kernel is $C_P(Q)$. It is straightforward to compute that $|N_P(Q)| = \frac{1}{2}p(p-1)$ and $C_P(Q) = Q$, so the image of ϕ is the unique subgroup of index 2 in the cyclic group $\mathrm{Aut}(Q)$.

By [Swa60, Lemma 1] (see also [AM94, Theorems 6.6 and 6.8]), the p -part of $H^k(P; \mathbb{Z})$ for $k \geq 1$ is the fixed points of the $N_P(Q)$ -action on $H^k(\mathbb{Z}/p; \mathbb{Z})$. We have that $H_1(Q; \mathbb{Z}) \cong Q$ and $H_2(Q; \mathbb{Z})$ is trivial, so by the Universal Coefficient Theorem, $H^2(Q; \mathbb{Z}) \cong Q$. Thus the action of $\mathrm{Aut}(Q) \cong (\mathbb{Z}/p)^\times$ on $H^2(Q; \mathbb{Z}) \cong \mathbb{Z}/p$ is given by multiplication in $\mathbb{Z}/p$. Now, by naturality of the cup product, we see that the action on $H^{2k}(Q; \mathbb{Z})$ is the k th power of the action on $H^2(Q; \mathbb{Z})$. Hence, the degrees $k = 2\ell$ with $\frac{1}{2}(p-1) \nmid \ell$, where $H^k(P; \mathbb{Z})$ contains p -torsion, are exactly the degrees for which the action is trivial. In particular, for $p \geq 7$, by applying the Universal Coefficient Theorem, we see that the group $H_3(P; \mathbb{Z})$ has no elements of order p .

Let Y be an oriented circle bundle over a closed genus $g \geq 0$ surface of Euler number n . Consider the composite $\varphi : \pi_1(Y) \twoheadrightarrow \mathbb{Z}/n \twoheadrightarrow \mathbb{Z}/p \hookrightarrow P$ which is non-trivial on a generator of the centre of $\pi_1(Y)$. We claim the bordism class $[Y, \varphi]$ vanishes in $\Omega_3(BP) \cong H_3(P; \mathbb{Z})$. Indeed, since $H_3(\mathbb{Z}/p; \mathbb{Z}) \cong \mathbb{Z}/p$ and $H_3(P; \mathbb{Z})$ has no elements of order p we see that map $H_3(\mathbb{Z}/p; \mathbb{Z}) \rightarrow H_3(P; \mathbb{Z})$ is zero. The proposition follows from applying Lemma 2.2. $\square$

Corollary 4.3. *For any $n \in \mathbb{Z}$, there is an oriented closed simply connected 4-manifold X and surface Σ smoothly embedded in X such that $\Sigma \cdot \Sigma = n$ and $\pi_1(X - \Sigma)$ is a non-abelian finite simple group.*

Proof. After applying Proposition 4.1 and Proposition 4.2 we are left with the case that n is divisible only by 2, 3, or 5. In these cases we use homomorphisms to the Lyons group Ly given, respectively, by $\pi_1(Y) \twoheadrightarrow \mathbb{Z}/2 \hookrightarrow Ly$, $\pi_1(Y) \twoheadrightarrow \mathbb{Z}/3 \hookrightarrow Ly$, or $\pi_1(Y) \twoheadrightarrow \mathbb{Z}/5 \hookrightarrow Ly$ which are all seen to vanish on third homology since $H_3(Ly; \mathbb{Z}) = 0$; see Table 1. $\square$

A classification? Whilst we have provided many examples of finite simple groups which can appear as complements of surfaces in simply connected

4-manifolds, we fall a long way short of a complete classification. To this end we raise the following question.

Question 4.4. Let G be a finite simple group. Which $(g, n) \in \mathbb{N}_{\geq 1} \times \mathbb{Z}$ ensure there exists a simply connected 4-manifold and $\Sigma_g \subset X$, a smoothly embedded surface of genus g , with $\Sigma_g \cdot \Sigma_g = n$ such that $\pi_1(X - \Sigma) \cong G$?

5. THE NON-PRIMITIVE CASE

The arguments and examples in the preceding sections produce surface complements $W = X - \Sigma$ with $H_1(W) = 0$ but for which $\pi_1(W)$ is non-trivial, answering the question of Kronheimer. For non-primitive classes (to which his question does not apply) the corresponding statement would be that there are surface complements for which the fundamental group of the complement is not the cyclic group $\mathbb{Z}/d$, where d is the multiplicity (defined in Remark 1.1) of the homology class $[\Sigma] \in H_2(X)$. In this short section we show that this holds by presenting examples of surfaces whose complements have non-abelian fundamental groups. The construction is explicit and does not require any of our group-theoretic arguments.

Proposition 5.1. *For any non-zero integers m and d , there is a simply connected 4-manifold containing a smoothly embedded surface of multiplicity d and self-intersection d^2m whose complement has non-abelian fundamental group.*

Proof. Recall Zeeman's d -twist spinning construction [Zee65], which from a knot K in S^3 produces a fibred knot $\tau_d(K)$ in S^4 . The fibre is $\Sigma_d(K)$, the d -fold branched cyclic cover of S^3 branched along K , minus a ball, and the monodromy is a generator of the covering transformations.

Write $n = d^2m$, and assume without loss of generality (as in Remark 1.1) that $m > 0$. The first step is to find a surface Σ_0 embedded in a simply connected manifold X with simply connected complement, with $\Sigma_0 \cdot \Sigma_0 = m$. For instance, one could take an algebraic curve in $\mathbb{C}P^2$ of sufficiently high degree so that its self-intersection is greater than m , and then blow up enough points to lower the self-intersection to m . Let X be the resulting blow-up of $\mathbb{C}P^2$. Note that the blown-up curve now meets a 2-sphere (any of the exceptional curves in the blow-up) in a single point and hence has simply connected complement. Then d times the homology class of Σ_0 is represented by a smoothly embedded surface $\Sigma_1 \subset X$ with $\pi_1(X - \Sigma_1) \cong \mathbb{Z}/d$.

Now replace Σ_1 by $\Sigma = \Sigma_1 \# \tau_d(K)$ where K is any non-trivial knot in S^3 ; we will show that the fundamental group of the complement of Σ is non-abelian. It is argued in [Kim06] that the fundamental group of $X - \Sigma$ is $\pi_1(S^3 - K)/\mu^d$, and that this contains the fundamental group of the d -fold cyclic cover of S^3 branched along K as an index d subgroup. By the solution of the generalized Smith conjecture [MB84, page 4], the branched cover of S^3 along a non-trivial knot cannot have trivial fundamental group. (This also follows from the orbifold theorem [CHK00, BP01, BLP05].) In other words, $\pi_1(S^3 - K)/\mu^d$ has a non-trivial index d subgroup. If $\pi_1(S^3 - K)/\mu^d$ were abelian, it would have to be isomorphic to its abelianization, $\mathbb{Z}/d$, and hence the only index d subgroup it contains would be trivial. It follows

that $\pi_1(S^3 - K)/\mu^d$, and hence the fundamental group of $X - \Sigma$, is non-abelian. $\square$

REFERENCES

- [AH61] M. F. Atiyah and F. Hirzebruch. Vector bundles and homogeneous spaces. In *Proc. Sympos. Pure Math., Vol. III*, pages 7–38. Amer. Math. Soc., Providence, RI, 1961. Cited on Page 3.
- [AM94] Alejandro Adem and R. James Milgram. *Cohomology of finite groups*, volume 309 of *Grundlehren der mathematischen Wissenschaften [Fundamental Principles of Mathematical Sciences]*. Springer-Verlag, Berlin, 1994. [10.1007/978-3-662-06282-1](#). Cited on Page 8.
- [Ati61] M. F. Atiyah. Bordism and cobordism. *Proc. Cambridge Philos. Soc.*, 57:200–208, 1961. [10.1017/s0305004100035064](#). Cited on Page 2.
- [BLP05] Michel Boileau, Bernhard Leeb, and Joan Porti. Geometrization of 3-dimensional orbifolds. *Ann. of Math. (2)*, 162(1):195–290, 2005. [10.4007/annals.2005.162.195](#). Cited on Page 9.
- [BP01] Michel Boileau and Joan Porti. Geometrization of 3-orbifolds of cyclic type. *Astérisque*, No. 272:208 pages, 2001. Appendix A by Michael Heusener and Porti. Cited on Page 9.
- [Bro87] Kenneth S. Brown. Finiteness properties of groups. In *Proceedings of the North-western conference on cohomology of groups (Evanston, Ill., 1985)*, volume 44 (1-3), pages 45–75, 1987. [10.1016/0022-4049\(87\)90015-6](#). Cited on Page 5.
- [Bro94] Kenneth S. Brown. *Cohomology of groups*, volume 87 of *Graduate Texts in Mathematics*. Springer-Verlag, New York, 1994. Corrected reprint of the 1982 original. Cited on Pages 3 and 5.
- [CCN⁺85] J. H. Conway, R. T. Curtis, S. P. Norton, R. A. Parker, and R. A. Wilson. *ATLAS of finite groups*. Oxford University Press, Eynsham, 1985. Maximal subgroups and ordinary characters for simple groups, With computational assistance from J. G. Thackray. Cited on Pages 6 and 7.
- [CHK00] Daryl Cooper, Craig D. Hodgson, and Steven P. Kerckhoff. *Three-dimensional orbifolds and cone-manifolds*, volume 5 of *MSJ Memoirs*. Mathematical Society of Japan, Tokyo, 2000. With a postface by Sadayoshi Kojima. Cited on Page 9.
- [DSE09] Mathieu Dutour Sikirić and Graham Ellis. Wythoff polytopes and low-dimensional homology of Mathieu groups. *J. Algebra*, 322(11):4143–4150, 2009. [10.1016/j.jalgebra.2009.09.031](#). Cited on Page 7.
- [Gor86] C. McA. Gordon. On the G -signature theorem in dimension four. In *À la recherche de la topologie perdue*, volume 62 of *Progr. Math.*, pages 159–180. Birkhäuser Boston, Boston, MA, 1986. Cited on Page 3.
- [Hem87] John Hempel. Residual finiteness for 3-manifolds. In *Combinatorial group theory and topology (Alta, Utah, 1984)*, volume 111 of *Ann. of Math. Stud.*, pages 379–396. Princeton Univ. Press, Princeton, NJ, 1987. Cited on Page 5.
- [Hig74] Graham Higman. *Finitely presented infinite simple groups*, volume No. 8 of *Notes on Pure Mathematics*. Australian National University, Department of Pure Mathematics, Department of Mathematics, I.A.S., Canberra, 1974. Cited on Page 5.
- [HS71] W. C. Hsiang and R. H. Szczarba. On embedding surfaces in four-manifolds. In *Algebraic topology (Proc. Sympos. Pure Math., Vol. XXII, Univ. Wisconsin, Madison, Wis., 1970)*, volume Vol. XXII of *Proc. Sympos. Pure Math.*, pages 97–103. Amer. Math. Soc., Providence, RI, 1971. Cited on Page 2.
- [JFT19] Theo Johnson-Freyd and David Treumann. Third homology of some sporadic finite groups. *SIGMA Symmetry Integrability Geom. Methods Appl.*, 15:Paper No. 059, 38, 2019. [10.3842/SIGMA.2019.059](#). Cited on Page 7.
- [Kim06] Hee Jung Kim. Modifying surfaces in 4-manifolds by twist spinning. *Geom. Topol.*, 10:27–56, 2006. [10.2140/gt.2006.10.27](#). Cited on Page 9.

- [Kir97] R.C. Kirby. Problems in low-dimensional topology. In W. Kazez, editor, *Geometric Topology*. American Math. Soc./International Press, Providence, 1997. Cited on Page 1.
- [KM93] P. B. Kronheimer and T. S. Mrowka. Gauge theory for embedded surfaces. I. *Topology*, 32(4):773–826, 1993. [10.1016/0040-9383\(93\)90051-V](#). Cited on Pages 1 and 6.
- [KM23] E. I. Khukhro and V. D. Mazurov. Unsolved problems in group theory. the Kourovka Notebook, 2023. [arXiv:1401.0300](#) [[math.GR](#)]. Cited on Page 1.
- [KR08] Hee Jung Kim and Daniel Ruberman. Smooth surfaces with non-simply-connected complements. *Algebr. Geom. Topol.*, 8(4):2263–2287, 2008. [10.2140/agt.2008.8.2263](#). Cited on Page 1.
- [Kre99] Matthias Kreck. Surgery and duality. *Ann. of Math. (2)*, 149(3):707–754, 1999. [10.2307/121071](#). Cited on Page 4.
- [Mal65] A. I. Mal'tsev. On the faithful representation of infinite groups by matrices. *Transl., Ser. 2, Am. Math. Soc.*, 45:1–18, 1965. [10.1090/trans2/045/01](#). Cited on Page 6.
- [MB84] John W. Morgan and Hyman Bass, editors. *The Smith conjecture*, volume 112 of *Pure and Applied Mathematics*. Academic Press, Inc., Orlando, FL, 1984. Papers presented at the symposium held at Columbia University, New York, 1979. Cited on Page 9.
- [McC01] John McCleary. *A user's guide to spectral sequences*, volume 58 of *Cambridge Studies in Advanced Mathematics*. Cambridge University Press, Cambridge, second edition, 2001. Cited on Page 3.
- [SW19] Markus Szymik and Nathalie Wahl. The homology of the Higman-Thompson groups. *Invent. Math.*, 216(2):445–518, 2019. [10.1007/s00222-018-00848-z](#). Cited on Page 5.
- [Swa60] Richard G. Swan. The p -period of a finite group. *Illinois J. Math.*, 4:341–346, 1960. Cited on Page 8.
- [Wal65] C. T. C. Wall. Finiteness conditions for CW-complexes. *Ann. of Math. (2)*, 81:56–69, 1965. [10.2307/1970382](#). Cited on Page 5.
- [Wal66] C. T. C. Wall. Surgery of non-simply-connected manifolds. *Ann. of Math. (2)*, 84:217–276, 1966. [10.2307/1970519](#). Cited on Page 3.
- [Wal99] C. T. C. Wall. *Surgery on compact manifolds*, volume 69 of *Mathematical Surveys and Monographs*. American Mathematical Society, Providence, RI, second edition, 1999. [10.1090/surv/069](#), Edited and with a foreword by A. A. Ranicki. Cited on Page 3.
- [Zee65] E. C. Zeeman. Twisting spun knots. *Trans. Amer. Math. Soc.*, 115:471–495, 1965. [10.2307/1994281](#). Cited on Page 9.